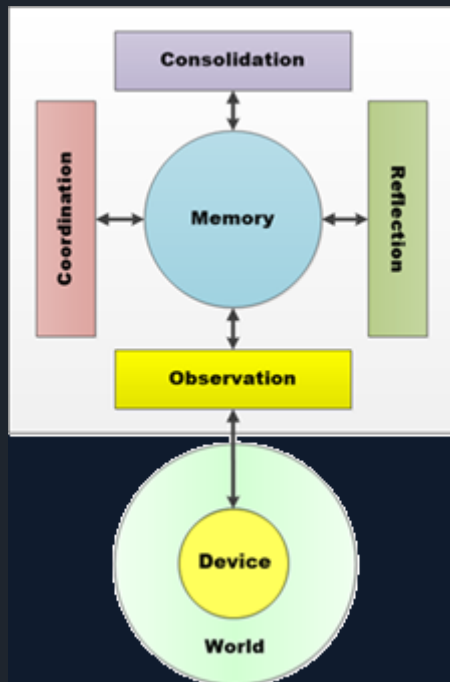


Self Improving Learner



Overview

01 Building Sentient Beings

The paper that defines the vision

02 GIL — Generally Intelligent Learner

Cognitive architecture with 14 design patterns

03 SIA — Self-Improving AI

The current LLM-based improvement loop

04 SIL — Self-Improving Learner

The merger of GIL and SIA

05 Premise Language

The platform that makes it possible

06 Benefits & Roadmap

What SIL adds and how to get there

Building Sentient Beings

Miller & Blumberg, 2025

<https://zenodo.org/records/15522356>

A sentient being is:

- **Embodied**
Has a body situated in a world
- **Perceiving**
Builds world model from direct sensation
- **Emotional**
Self-regulates via valence and arousal
- **Situationally Aware**
Discerns and predicts elements in its vicinity
- **Autonomous**
Mission-oriented, not task-oriented

Memory as Totality

Perceptual

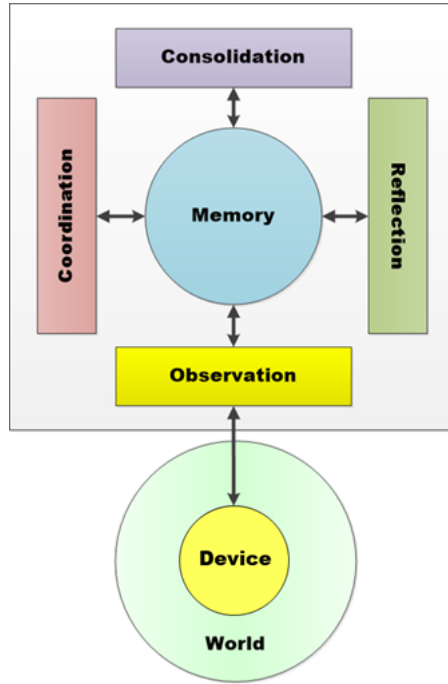
Associative

Imaginative

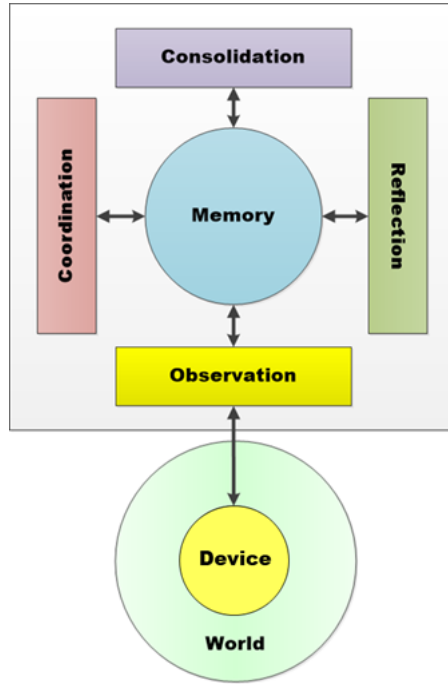
Ontological

Activative

GIL — Generally Intelligent Learner



GIL — Generally Intelligent Learner



<https://zenodo.org/records/15522356>

GIL — Generally Intelligent Learner



Coordination

Activate · Associate
Solve · React · Deliberate



Consolidation

Forget · Automate
Compress



Reflection

Motivate · Regulate
Compensate · Explore



Observation

Perceive · Detect
Store · Retrieve · Match

T O T A L I T Y
Shared Knowledge Base

GIL's 14+ Agent Design Patterns

*Cognivite architecture agent design patterns, **Building Minds with Patterns***

Observation

Mind/Body

Observation

Reminding

Coordination

Coordination

Reaction

Deliberation

Reflection

Motivation

Simulation

Regulation

Compensation

Coping

Exploration

Discovery

Imitation

Play

Meta-Control

Consolidation

Consolidation

GIL's Memory Areas

Imaginative

Canvas · Taxis · Animation

Mental theater for simulation

Ontological

Schemes · Beliefs · Lexemes

Associative

Cases · Traits · Matcher (Jaccard)

The concordance of experience

Perceptual

Percepts · Features · Image Schemas

Raw and parsed sensory elements

Control

Registry · Actor · Codes · Loop

Activative

Realities · Glue · Viewpoints · Activation

What is relevant right now

SIA — Self-Improving AI - HEXO Labs

Three-agent self-improvement loop

Meta-Agent

Creates initial
target_agent.py from
task description

Target Agent

Executes the task
Produces results
(Python subprocess)

Feedback Agent

Reviews results
Rewrites improved
target_agent.py

Generation Loop: Gen 1 → Gen 2 → Gen 3 → ... → Gen N (each rewrites target_agent.py)

SIA by the numbers

4,143

Lines of Python

20

Source files

3

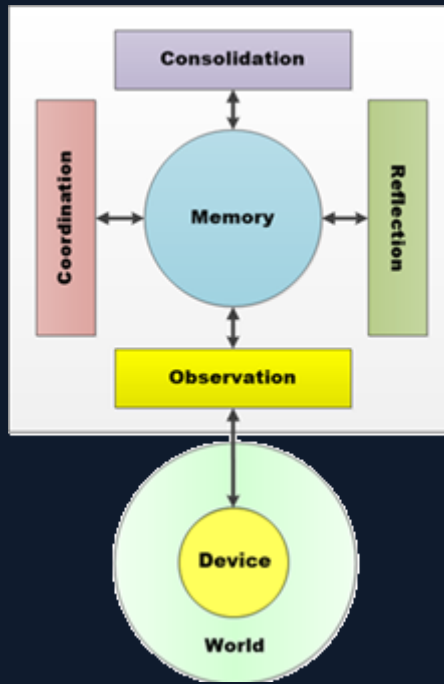
Agent impls

6

LLM providers

SLA Potential Improvements

- ❌ **Cross-Run Memory**
Every run starts from scratch. No persistent learning.
- ❌ **Emotional Regulation**
No frustration detection, no self-regulation, no coping strategies.
- ❌ **Consolidation**
No forgetting, no compression, no skill formation from experience.
- ❌ **Transparent Reasoning**
LLM reasoning is implicit, unrepeatable, and forgotten.
- ❌ **Forward Models**
Cannot predict outcomes before executing. Only reacts to results.
- ❌ **Trait-Based Retrieval**
Cannot search for similar prior tasks. No associative memory.
- ❌ **Autonomous Motivation**
Goals come from task.md, never self-generated.
- ❌ **Mental Simulation**
Cannot imagine alternatives before committing to one.



Self-Improving Learner (SIL)

Merging Cognitive Architecture with Self-Improving AI

Hexo Labs · SubThought Corporation · Michael S. P. Miller · June 2026

Confidential

SIL — Self-Improving Learner



SIL — Self-Improving Learner



SIL — Self-Improving Learner

GIL's cognitive architecture + SIA's computational domain



SIL — Self-Improving Learner

GIL's cognitive architecture + SIA's computational domain

Same Mind. Different Body.



SIL — Self-Improving Learner

GIL's cognitive architecture + SIA's computational domain

Same Mind. Different Body.

GIL provides

- 14 design patterns
- 5 memory areas
- Equilibration cycle
- Activation spreading
- Emotional regulation



SIL — Self-Improving Learner

GIL's cognitive architecture + SIA's computational domain

Same Mind. Different Body.

GIL provides

- 14 design patterns
- 5 memory areas
- Equilibration cycle
- Activation spreading
- Emotional regulation

Premise provides

- Persistent KB
- HTTP + JSON native
- Shell execution (%)
- Concurrent agents
- FIPA-ACL networking



SIL — Self-Improving Learner

GIL's cognitive architecture + SIA's computational domain

Same Mind. Different Body.

GIL provides

- 14 design patterns
- 5 memory areas
- Equilibration cycle
- Activation spreading
- Emotional regulation

Premise provides

- Persistent KB
- HTTP + JSON native
- Shell execution (%)
- Concurrent agents
- FIPA-ACL networking

SIA provides

- Python runtime (body)
- LLM API access
- Benchmark tasks
- Evaluation pipeline
- File system I/O



SIL Architecture

MIND (Premise / GIL)

Observation

Perceiver
Detectors
Storer/Matcher

Coordination

Activators
Solvers
Reactor/Deliberator

Reflection

Correlator
Regulator
Compensator

Consolidation

Amneator
Automator
Compressor

T O T A L I T Y

BODY (Python / SIA)

Psyche App

File System

Shell (subprocess)

LLM APIs

Venv / pip

Evaluation Scripts

Homeostatic Needs

Accuracy · API-Budget · Time · Disk

12 Actuations: run-script · read-file · write-file · make-dir · copy-file · delete-file · install-package · create-venv · prompt-llm · to-json · from-json · read-env

8 Realities: Observed · Desired · Expected · Possible · Attained · Confirmed · Imagined · Recalled

No generation loop — four components run concurrently. System stops on homeostatic equilibrium.

Action & Reason Library

Replaces opaque Python files with queryable, composable, regulatable knowledge

Action

:Context	unison of preconditions
:Goals	series of actuations/sub-actions
:Loop	iteration control
:Result	unison of postconditions
:Reliability	success rate -1..1

Reason

:Premises	unison of preconditions
:Known	unison of background knowledge
:Lemmas	unison of bridging reasons
:Proof	series of logical steps
:Theorem	unison of conclusions

Action & Reason Library

Replaces opaque Python files with queryable, composable, regulatable knowledge

Action

:Context	unison of preconditions
:Goals	series of actuations/sub-actions
:Loop	iteration control
:Result	unison of postconditions
:Reliability	success rate -1..1

Reason

:Premises	unison of preconditions
:Known	unison of background knowledge
:Lemmas	unison of bridging reasons
:Proof	series of logical steps
:Theorem	unison of conclusions

Why this matters

Queryable

"Which Actions produce submission.csv?"

Composable

Actions reference other Actions as steps

Regulatable

Modify individual failing steps, not entire programs

Transferable

Strategies carry across tasks via trait matching

SIL vs SIA — What's Added

Capability	SIA	SIL
Cross-run memory	None — every run starts fresh	Persistent Totality across all runs
Forward model	None — only reacts to results	Expected reality predicts outcomes
Strategy retrieval	None — LLM reinvents each time	Matcher finds Cases by Jaccard similarity
Emotional regulation	None — no self-regulation	Valence/Arousal drive coping strategies
Consolidation	None — append-only context	Forget, compress, automate skills
Mental simulation	None — commits blindly	Canvases explore alternatives first
Justification	Lost in LLM context window	Reasons persist with full Proof chains
Goal management	Hardcoded in task.md	Homeostatic needs create Objectives



The Premise Language

The platform that makes SIL possible



Persistent KB with pattern matching



Concurrent agents (services, agents, rules)



HTTP client (tell/ask) with JSON
encode/decode



Shell execution (%) with structured output



4-layer Rete inference engine



FIPA-ACL inter-agency communication



The Premise Language

The platform that makes SIL possible

<https://github.com/subthought/premise>

How Premise Aids the Rewrite

context_manager.py
558 lines of file I/O



KB IS the context
~30 lines of queries

KB replaces files

agent_impls/ (4 files)
421 lines of SDK wrappers



Native ask/tell
~20 lines total

Network is built-in

profiles.py + providers.py
192 lines JSON loading



KB relations
~30 lines

Config is KB data

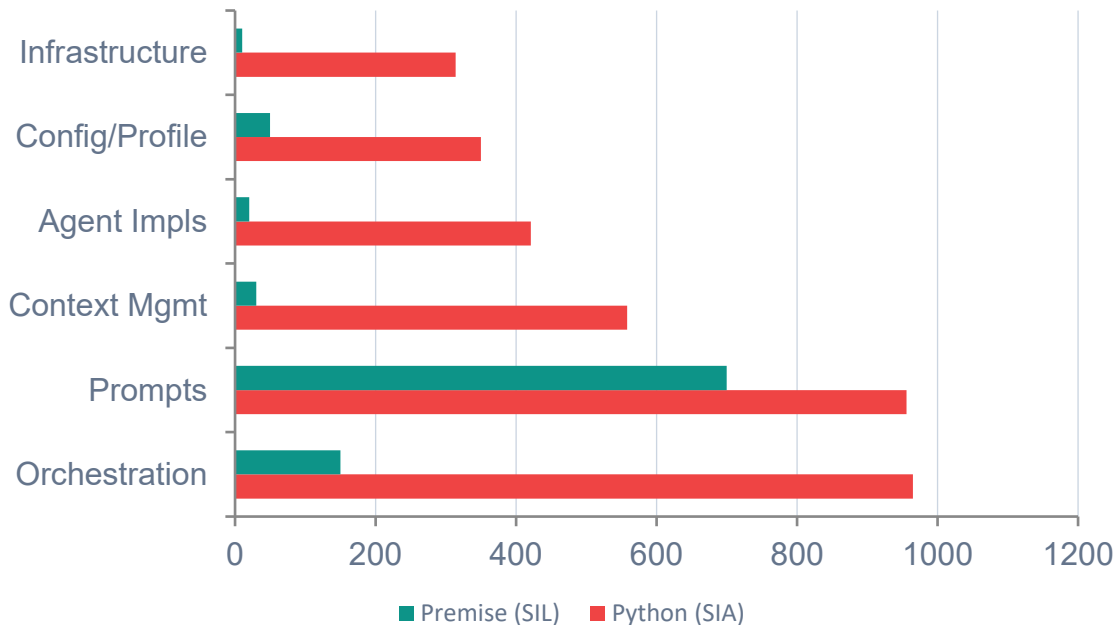
prompts.py
956 lines of templates



Task perceived, not
described in English

**Perceive, don't
prompt**

Code Reduction: Python → Premise



75%
reduction

4,143 → ~1,050
lines of code

KB eliminates
serialization

Native networking
eliminates SDK wrappers

Concise syntax
eliminates boilerplate

Implementation Roadmap

1

SIL Totality

Adapt GIL's knowledge base for computational domain. Rescale potentiation windows. Define modalities, codes, and image schemas.

3

Observation

Perceiver, 6 detectors, Storer/Retriever/Matcher. Trait-based case indexing.

5

Reflection

Ameliorator, Correlator, Regulator, Compensator, Simulator. Self-improvement.

7

Bootstrap & Test

Run against LawBench. Verify equilibration discovers TF-IDF strategy.

2

Psyche App

Python firmware (~100 lines). Registers Device, polls filesystem, dispatches actuations.

4

Coordination

Activators, case-based Solver, Reactor, Deliberator. Action selection from KB.

6

Consolidation

Amneator, Automator, Compressor. Forgetting, skill formation, compression.

8

Validate Transfer

Cross-task (chess after LawBench) and cross-run (LawBench run 2) memory.



SIL doesn't just improve code. *It understands why.*

SIA uses LLMs for both reasoning and execution.
SIL uses GIL's mechanisms for reasoning and LLMs for execution.
The mind is GIL. The LLM is a tool the mind uses.

HEXO Labs · SubThought Corporation

Contact: SubThought@hotmail.com